

Building 207 – Split-Air AHUs Sequence of Operation

Run Conditions - Requested:

The unit shall operate when:

- Scheduled to run based on a time of day schedule or override schedule.
- OR unit control is overridden.

Freeze Protection:

The unit shall generate an alarm upon receiving a low temperature detection status. The OA and EA dampers shall close and the RA damper shall open. The two chilled water valves shall be commanded closed and the two hot water valves shall be modulated to maintain a 200°F (adj.) setpoint at both the MA/OA and MA/RA temperature sensors. There is no automatic fail-safe position on the valve actuators as the equipment is on emergency power.

RA Smoke Detection:

The unit shall shut down and generate an alarm upon receiving a RA smoke detector status.

SA Smoke Detection:

The unit shall shut down and generate an alarm upon receiving a SA smoke detector status.

Optimal Start:

The unit shall start prior to scheduled occupancy based on the time necessary for the zones to reach their occupied setpoints. The start time shall automatically adjust based on changes in OA temperature and zone temperatures. Minimum OA control shall be disabled. Economizer mode is permitted.

SA and RA Fan:

The SA and RA fan shall run anytime the unit is commanded to run, unless shutdown on safeties. The VFDs on the fans are hardwire-interlocked to the smoke detectors, high static pressure cutouts, and LTDE in the OA duct. As long as the above safeties

are closed and the VFDs are in the auto position, the unit shall be enabled to start. When the unit is commanded to run, EP-1 will be energized to provide main air to supply air and return air smoke dampers D-4 and D-5 so that they open. This does not apply to AHUs 1-2, 3-5, 4-1, and 4-2 that don't have smoke dampers. Once the end switches (ES-1 and ES-2) on the smoke dampers are enabled, relay two (RE-2) shall be energized and the VFD will be allowed to start. ~~As long as all safeties are closed, relay one (RE-1) will remain energized and provide power to EP-2 so that main air is provided to the OA/RA/EA damper actuators pilot positioners.~~ The SA and RA fan will start and shall be ramped up to the static pressure setpoint over five minutes (adj.). To prevent short cycling, the SA fan and RA fan shall have a minimum runtime of 5 minutes (adj.).

Pneumatic damper actuators are replaced with electronic Belimo Actuators

When the SA fan is in alarm, the chilled water valve and hot water valve on the AHU shall be controlled according to the freeze protection mode of operation stated above as long as the OA temperature is below 55°F (adj.). If the OA Temperature is above 55°F (adj.) then the hot water valve will fail closed.

Alarms shall be provided as follows:

- SA Fan Failure or locally switched Off: Commanded on, but the status is off.
- SA Fan in Hand: Commanded off, but the status is on.

SA Duct Static Pressure Control:

The controller shall measure duct static pressure as sensed by PTE-1 and modulate the SA fan VFD speed to maintain a duct static pressure setpoint. The speed shall not drop below 30% (adj.).

- The initial duct static pressure setpoint shall be 1.0 in H₂O (adj.).

Alarms shall be provided as follows:

- High SA Static Pressure: If the SA static pressure is 25% (adj.) greater than setpoint.
- Low SA Static Pressure: If the SA static pressure is 25% (adj.) less than setpoint.

RA Fan:

Alarms shall be provided as follows:

- RA Fan Failure or locally switched Off: Commanded on, but the status is off.
- RA Fan in Hand: Commanded off, but the status is on.

RA Fan Tracking:

The return air flow setpoint shall be determined by measuring the supply air flow as

measured by VPT-1 and subtracting the fixed outside air flow rate as determined for each AHU. Flow measuring station VPT-2 will provide the return air flow reading for each AHU. See table below for outside air flow rates:

AHU	Outside Air Flow Rate
1-2	7629
1-4	1751
2-1	2837
2-2	600
2-4	3900
2-5	1700
2-6	3788
3-1	500
3-2	788
3-4	4074
3-5	830
4-1	1314
4-2	3468

Alarms shall be provided as follows:

- High RA Flow Rate: If the RA flow rate is 25% (adj.) greater than setpoint.
- Low RA Flow Rate: If the RA flow rate is 25% (adj.) less than setpoint.

Heating Coil Valve in the MA/OA Duct:

The controller shall measure the SA temperature and modulate the MA/OA heating coil valve to maintain the SA temperature setpoint. When the OA temperature falls below 38°F (adj.) the MA/OA heating coil valve shall be controlled by the MA/OA temperature sensor to maintain a setpoint of 40°F (adj.).

The MA/OA heating valve shall be enabled whenever:

- OA temperature is less than 45°F (adj.).
- AND the economizer is disabled.
- AND the SA fan status is on.
- AND the cooling is not active.

The MA/OA heating coil valve shall control based upon the freeze protection sequence whenever:

- The Low Temperature Detector Electric (LTDE) is on. Typically set at 36°F.

Alarms shall be provided as follows:

- Low SA Temp: If the SA temperature is 5°F (adj.) less than setpoint.

Heating Coil Valve in the MA/RA Duct:

The controller shall measure the SA temperature and modulate the MA/RA heating coil valve to maintain the SA temperature setpoint only when the MA/OA heating valve cannot maintain supply air temperature.

The MA/RA heating valve shall be enabled whenever:

- OA temperature is less than 45°F (adj.).
- AND the economizer is disabled.
- AND the SA fan status is on.
- AND the cooling is not active.
- AND the MA/OA heating valve is fully open.

The MA/RA heating coil valve shall control based upon the freeze protection sequence whenever:

- The Low Temperature Detector Electric is on. Typically set at 36°F.

Alarms shall be provided as follows:

- Low SA Temp: If the SA temperature is 5°F (adj.) less than setpoint.

Cooling Coil Valve in the MA/OA Duct:

The controller shall measure the MA/OA temperature and modulate the MA/OA cooling coil valve to maintain MA/OA temperature setpoint of 55°F (adj.).

The MA/OA cooling valve shall be enabled whenever:

- OA temperature is greater than 55F (adj.).
- AND the economizer is disabled or fully open.
- AND the SA fan status is on.
- AND the heating is not active.

The MA/OA cooling coil valve shall close whenever the Low Temperature Detector Electric is on.

Alarms shall be provided as follows:

- High MA/OA Temp: If the MA/OA temperature is 5°F (adj.) greater than setpoint.

Cooling Coil Valve in the MA/RA Duct:

The controller shall measure the SA temperature and modulate the MA/RA cooling coil valve to maintain SA temperature setpoint.

The MA/RA cooling valve shall be enabled whenever:

- OA temperature is 10°F (adj.) greater than OA temperature setpoint for the MA/OA cooling valve.
- AND the economizer is disabled.
- AND the SA fan status is on.
- AND the heating is not active.
- OR the outside air temperature is below 38°F (adj.).

The MA/RA cooling coil valve shall close whenever the Low Temperature Detector Electric is on.

Alarms shall be provided as follows:

- High SA Temp: If the SA temperature is 5°F (adj.) greater than setpoint.

Damper Operation

When the unit is commanded to run and it is not in the economizer mode of operation, the OA and RA damper shall be command 100% open and the EA damper shall be 100% closed. However, when the outside air temperature is below 38°F (adj.) when an AHU is started in any mode, the OA damper shall be kept closed for the ramp time static pressure control startup of the VFD's and the MA/OA hot water valve will be commanded 100% open for the ramp time static pressure control startup of the VFD's plus one minute (adj.). Once this startup routine is complete, the MA/OA hot water valve will be allowed to slowly close over a five minute period (adj.), i.e. a deceleration mode, before it is released to normal operational control.

Economizer:

The controller shall measure the MA/OA temperature and command the OA damper to 100% open and modulate the RA/EA dampers in sequence to maintain a setpoint of 2°F (adj.) less than the SA temperature setpoint.

The economizer shall be enabled whenever:

- The outside air enthalpy is less than the return air enthalpy.
- AND the SA fan status is on.
- AND the outside air temperature is above 38°F (adj.). **Adjusted to 43°F**

The dampers shall go to there normal positions whenever:

- The Low Temperature Detector Electric is on.
- OR on loss of SA status.

The OA and EA dampers shall close and the RA damper shall open when the unit is off for any reason, and when the unit is running after hours in an unoccupied mode.

Dehumidification:

The controller shall measure the RA humidity and override the cooling sequence if the RA humidity is equal to or greater than sixty percent (60%) (adj.). The supply air temperature setpoint shall be 55°F. Dehumidification shall be enabled whenever the SA status is on. Dehumidification will be disabled when the RA humidity is two percent (2%) (adj.) less than the high humidity limit setpoint.

Humidification: Disabled

The controller shall measure the RA humidity to determine operation of the humidifier pump. When the humidity is greater than 38% (adj.) the humidifier will be turned off.

Humidification shall be enabled whenever:

- The SA fan status is on.
- AND the OA temperature is below 45°F (adj.).
- AND the RA humidity is less than 35% (adj.).

MA Temperature:

Alarms shall be provided as follows (Always active):

- High MA Temp: If the MA/OA and MA/RA temperature is greater than 90°F (adj.).
- Low MA Temp: If the MA/OA and MA/RA temperature is less than 36°F (adj.).

RA Humidity:

Calculate and display absolute humidity on the graphic screen.

Alarms shall be provided as follows:

- High RA Humidity: If the RA humidity is greater than 70% (adj.).
- Low RA Humidity: If the RA humidity is less than 25% (adj.).

RA Temperature:

Alarms shall be provided as follows:

- High RA Temp: If the RA temperature is greater than 90°F (adj.).
- Low RA Temp: If the RA temperature is less than 45°F (adj.).

SA Temperature:

Alarms shall be provided as follows:

- High SA Temp: If the SA temperature is greater than 105°F (adj.).
- Low SA Temp: If the SA temperature is less than 45°F (adj.).

SA Temperature Setpoint:

The SA temperature setpoint shall be 62°F (adj.) year round.

Unoccupied Mode:

The unit shall shutdown by occupancy schedule. The OA and EA dampers shall close and the RA damper shall open. The chilled water valves, hot water valves, and humidifier valve shall close.

END OF SEQUENCE